

EstatePulse CRM — Database & API Overview

Technical Interview Project Submission • Full-Stack Operations OS

Live Web Application	https://real-estate-crm-5k5j.onrender.com
Backend API Root	https://estatepulse-backend-56wz.onrender.com
Interactive Swagger Docs	https://estatepulse-backend-56wz.onrender.com/api/docs
GitHub Repository	https://github.com/Yogesh-kk49/Real_estate_CRM

1. Demo Evaluation Credentials

The application enforces strict Role-Based Access Control (RBAC). Reviewers can sign in with the following pre-configured credentials (or use the one-click modal loaders):

Role	Email	Password	Privileges & Scoping
Administrator	admin@coromandel.in	Admin@1234	Full organization oversight, recruit sales consultants, all leads, inventory control
Sales Consultant	meera@coromandel.in	Sales@1234	Scoped lead workspace (assigned leads only), customer timeline, unit booking
Sales Consultant	anand@coromandel.in	Sales@1234	Scoped lead workspace (assigned leads only), customer timeline, unit booking

2. Database Architecture (PostgreSQL 16)

The production system is powered by a **Managed PostgreSQL 16** cloud database on Render. SQLAlchemy 2.0 ORM manages relational transactions, connection pooling (pool_pre_ping=True, pool_recycle=300), foreign keys, and row-level locks.

Entity Table	Primary Key / Indexes	Key Relational Constraints	Business Purpose
users	id (PK) email (UK, Indexed)	role: ADMIN SALES_EMPLOYEE	User authentication, salted bcrypt hashes (72-byte safe limit), RBAC access levels.
projects	id (PK)	1:N with buildings	Chennai luxury developments (location, status, completion timeline).
buildings	id (PK) project_id (FK)	1:N with units	Building towers and wings within luxury developments.
units	id (PK) building_id (FK)	availability: Available Reserved Booked	Real-time inventory matrix: super built-up sq.ft, floor, facing, INR pricing.
leads	id (PK) assigned_user_id (FK)	7-Stage lifecycle state machine	Buyer lead portfolio, budget bounds (Lakhs/Crores), scheduled follow-up dates.
lead_notes	id (PK) lead_id (FK), author_id (FK)	Indexed on created_at	Chronological 360° interaction logs (Call, Meeting, WhatsApp, Site Visit).
bookings	id (PK) unit_id (FK, UK)	UNIQUE(unit_id) constraint	Transaction ledger: token deposits, agreement values, payment references.

3. Concurrency Guard: Zero Double-Booking Architecture

To eliminate double-allocations when multiple agents book the same high-demand villa at the exact same millisecond, EstatePulse executes an **atomic conditional UPDATE statement** inside the database transaction:

```
# backend/app/services/booking_service.py
stmt = update(Unit).where(Unit.id == unit_id, Unit.availability ==
'Available').values(availability='Booked')
result = db.execute(stmt)
if result.rowcount == 0:
    db.rollback()
    raise HTTPException(status_code=409, detail='Unit was just booked by another user.')
```

- **First Request:** Finds availability = 'Available', updates row to Booked, returns **HTTP 201 Created**.
- **Second Concurrent Request:** Matches rowcount == 0, rolls back, returns **HTTP 409 Conflict**.
- **Secondary Safety:** Unique index constraint on bookings.unit_id.

4. Complete API Endpoints Specification

Base URL: <https://estatepulse-backend-56wz.onrender.com/api> • Bearer Token Auth via Authorization: Bearer <JWT>

Category	Method & Path	Access	Request Payload / Parameters	Status
Auth	POST /auth/login	Public	email, password	200, 401
Auth	GET /auth/me	Authenticated	Bearer token header	200, 401
Users	GET /users/employees	Authenticated	Active sales consultants	200
Users	GET /users	Admin Only	Team roster & lead stats	200, 403
Users	POST /users	Admin Only	Recruit staff (SALES_EMPLOYEE only)	201, 400
Users	PATCH /users/{id}	Admin Only	Update profile / toggle active	200, 404
Leads	GET /leads	RBAC Scoped	Query: stage, priority, search	200
Leads	POST /leads	Authenticated	name, email, phone, budget, date	201, 400
Leads	GET /leads/{id}	RBAC Scoped	Full 360° lead timeline dossier	200, 403
Leads	PATCH /leads/{id}	RBAC Scoped	Lifecycle stage transition	200, 400
Leads	POST /leads/{id}/notes	RBAC Scoped	note_type, touchpoint content	201
Leads	POST /leads/{id}/assign	Admin Only	assigned_user_id (Delegation)	200, 403
Properties	GET /properties/projects	Authenticated	Developments + inventory stats	200
Properties	GET /properties/units	Authenticated	Query: project_id, floor, availability	200
Bookings	POST /bookings	Authenticated	lead_id, unit_id, agreement, deposit	201, 409
Bookings	GET /bookings	RBAC Scoped	Booking agreements ledger	200
Dashboard	GET /dashboard/stats	RBAC Scoped	Pipeline KPIs, follow-up alerts, funnel	200

5. Technical Architecture Highlights

- **Automated Cloud Bootstrapping:** On initial boot against an empty PostgreSQL database, the backend automatically initializes schema tables and seeds default demo accounts so logins work immediately with zero manual shell commands.
- **Strict Role Escalation Guard:** System administrators can onboard sales consultants (SALES_EMPLOYEE), but backend Pydantic validators explicitly prohibit creating additional administrators to preserve organizational hierarchy.
- **Input Normalization & Validation:** Indian telephone patterns (+91 or 10-digit formats), future-dated follow-up dates, and Lakhs/Crores currency denominations are strictly validated.
- **Automated Test Suite Verified:** 100% test pass rate across concurrency race conditions, RBAC authorization, and data validation.